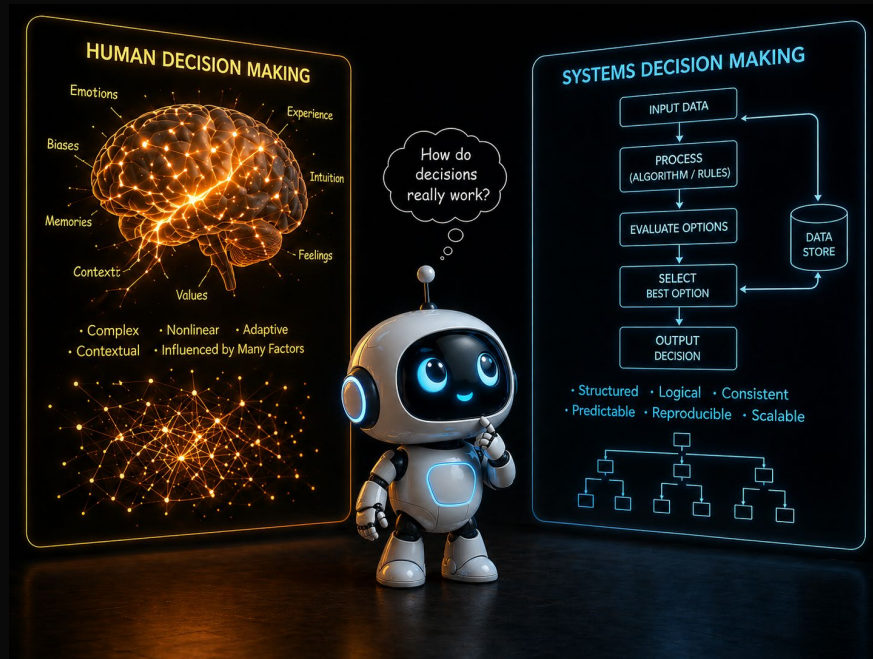




ITAI 4374

Module 08

Decision Making

From Brain to Autonomous Systems

How does the brain choose what to do,
and how do AI systems make decisions?



Where We Are in the Journey

- ✓ Module 02: COMPUTE What does the brain compute?
- ✓ Module 03: BUILD Brain anatomy and neurons
- ✓ Module 04: CONNECT Spiking neural networks
- ✓ Module 05: SENSE Sensory processing and perception
- ✓ Module 06: REMEMBER Memory and learning
- ✓ Module 07: ATTEND AND AWARE Attention and consciousness
- ▶ **Module 08: DECIDE** Decision making you are here

The Problem: Too Many Possible Actions

Every second, the brain could launch many possible actions.

Move left. Move right. Stand up. Sit down.
Speak. Stay silent. Approach. Retreat.

The decision problem is selection under uncertainty.

The brain must choose one action, suppress the rest, and commit fast enough to act.

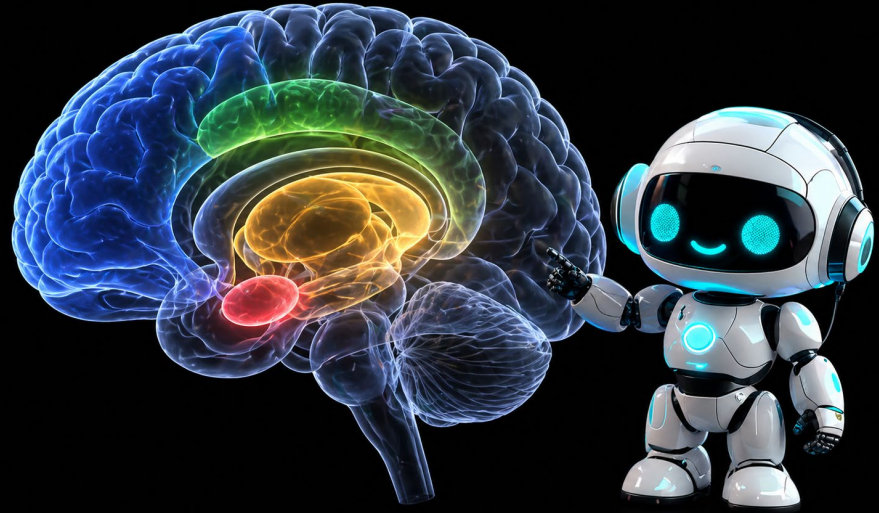




PART 1

The Brain's Decision System

How does the brain
choose what to do?



The Brain's Decision Team

No single brain region decides alone. Three core regions work together, with a fourth region monitoring the process.

Prefrontal Cortex

Plans, compares options, and keeps the goal in mind

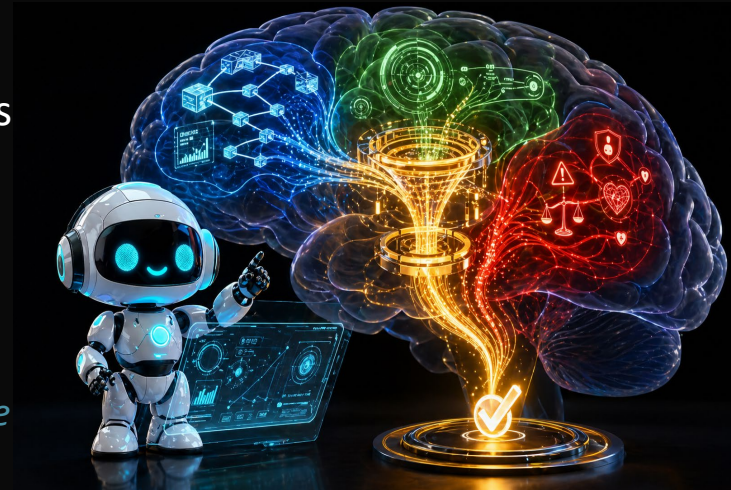
Basal Ganglia

Selects which action gets released and which get suppress

Amygdala

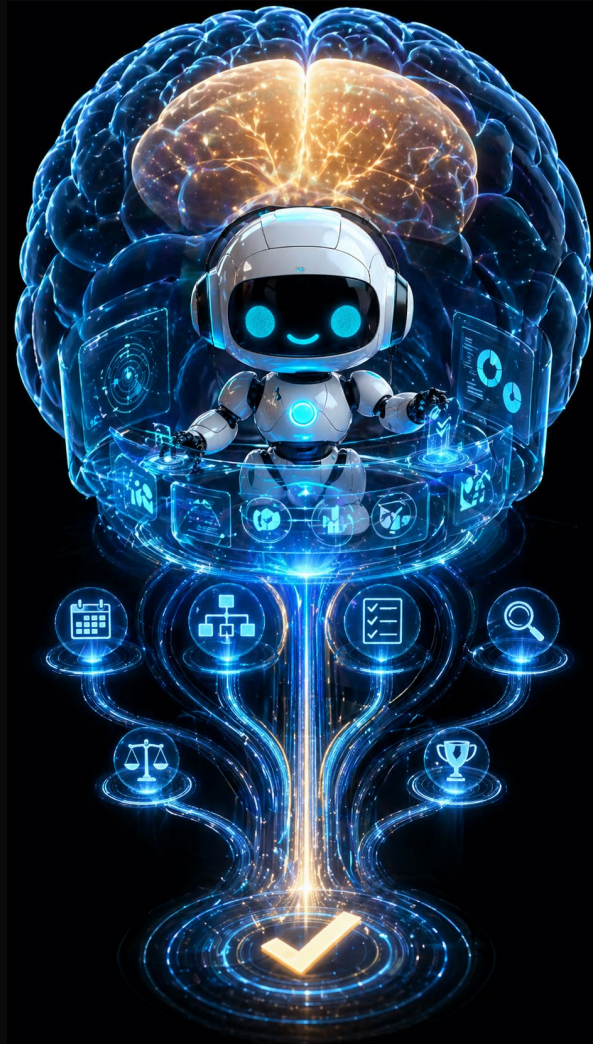
Adds emotional and risk value to each option

A fourth region, the Anterior Cingulate Cortex, monitors the entire process. We will meet it shortly.



Prefrontal Cortex: The Executive

- The prefrontal cortex is not one thing. Its subdivisions support different parts of decision making.
- **dIPFC**: working memory
 - Keeps options active while you compare them
- **vmPFC**: emotional integration
 - Connects options with personal meaning and emotional value
- **OFC**: value scoring
 - Estimates how rewarding or costly an option may be
- **AI parallel**: context windows and value functions



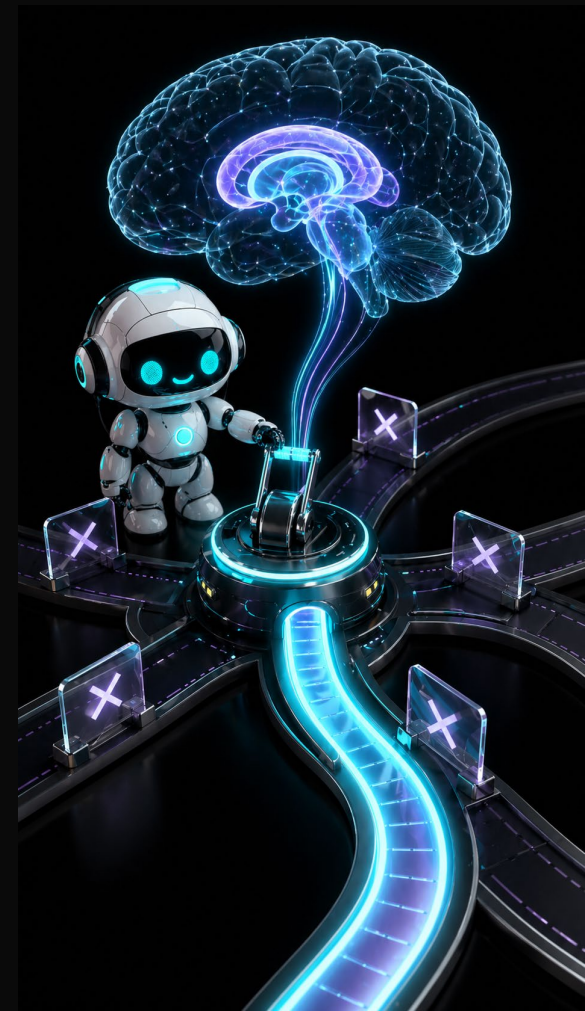
Why Emotion Is Part of Good Decisions

- **Damasio's somatic marker hypothesis:**
 - Past experiences attach body based emotional signals to future options.
- **Iowa Gambling Task:**
 - Healthy participants often avoid risky decks before they can explain why.
 - Their bodies respond first. Palms sweat before conscious awareness catches up.
- **Patients with vmPFC damage often fail to develop this early warning signal.**
 - They can reason logically, but still make poor real world choices.
- **AI gap:** options do not feel safe, risky, fair, or dangerous to an AI system.



Basal Ganglia: The Action Selector

- The brain does not release every possible action. It keeps most actions inhibited.
- The basal ganglia help decide what to release and what to suppress.
- Go pathway
 - Helps release the selected action
- No Go pathway
 - Helps suppress competing actions
- Dopamine changes the balance between Go and No Go.
 - This helps explain why Parkinson's disease can make movement hard to initiate.
- **AI parallel:** actor critic reinforcement learning

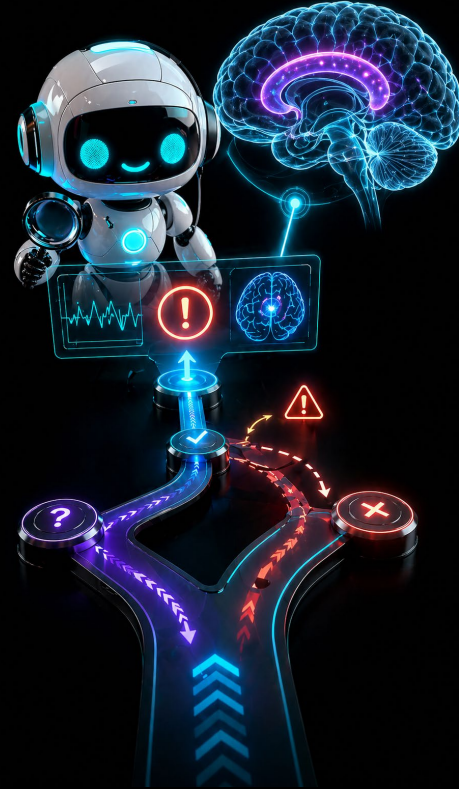


Anterior Cingulate Cortex: The Monitor

While the other regions decide and act, the ACC watches the entire process.

- **Conflict detection**
 - Notices when two options feel equally strong and signals that the choice is hard
- **Error detection**
 - Catches mistakes the moment they happen, often before the action is finished
- **Effort signaling**
 - Tells the rest of the brain when a task needs more focus and slower thinking

AI parallel: confidence scoring, uncertainty estimation, and self correction prompts



The Four Regions in One Decision

Scenario: you are about to reply to an angry email.

Amygdala

Flags threat and prepares an emotional response

Prefrontal Cortex

Considers the sender, the stakes, and the long term goal

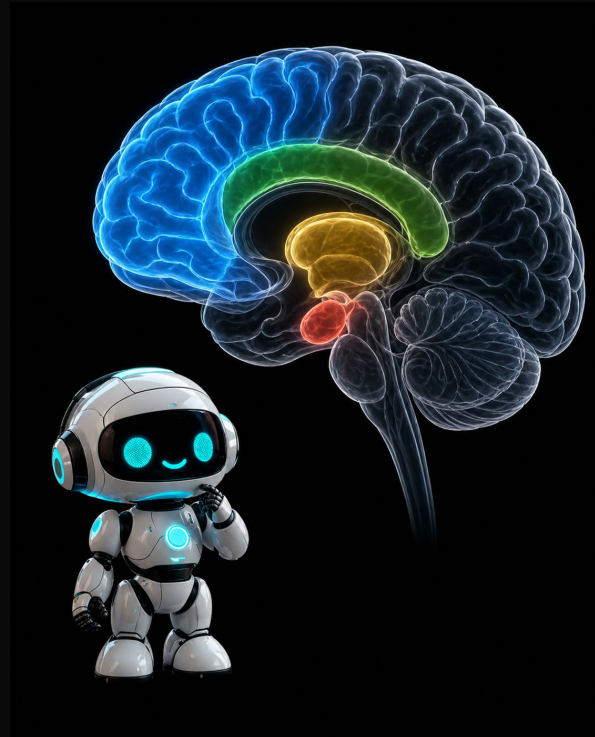
Basal Ganglia

Releases the calmer response and suppresses the impulsive one

ACC

Monitors conflict and catches possible mistakes while you type

The decision emerges from parallel activity, not a single command center.



Three Types of Decisions

- Different decisions rely on overlapping but different circuits.
- **Perceptual decisions**
 - Detect a signal in noise. Is that a car or a shadow?
 - Evidence builds until a threshold is reached.
- **Value based decisions**
 - Compare costs, benefits, and expected outcomes.
 - The OFC and vmPFC are especially important.
- **Social decisions**
 - Involve fairness, trust, punishment, cooperation, and reputation.
 - These require modeling other minds.

AI parallel: image classifiers (perceptual), recommendation systems (value), language models (social)



Social Decision Making: The Hardest Type

- Social choices require predictions about what other people think, want, and expect.
- **Ultimatum Game:**
 - Player A splits \$10. Player B can accept or reject.
 - A narrow reward maximizing strategy would accept any positive offer.
 - Humans often reject unfair offers to punish unfairness.
- The brain values fairness and social cost, even when money is lost.

AI gap:

- AI can imitate social reasoning from patterns in data, but it does not truly experience trust, fairness, reputation, or social obligation.



How the Brain Commits to a Choice

The brain does not simply flip a switch. It accumulates evidence over time.

Drift

How strongly the evidence favors one choice

Noise

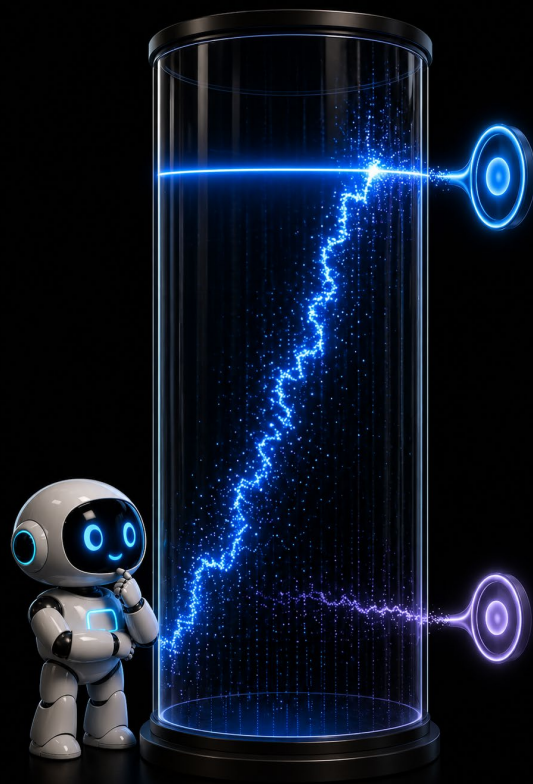
Random variation that can change errors and response times

Threshold

How much evidence is required before the system commits

The Drift Diffusion Model explains both the choice and the time it takes to make it.

AI parallel: confidence thresholds in classifiers



Speed vs Accuracy: A Universal Tradeoff

Every decision system faces the same tradeoff.

- Lower threshold
 - Faster decisions, more mistakes
- Higher threshold
 - Slower decisions, fewer mistakes
- The brain changes the threshold based on context.
 - High stakes or unfamiliar situations require more evidence.
 - Routine or time critical situations require faster commitment.

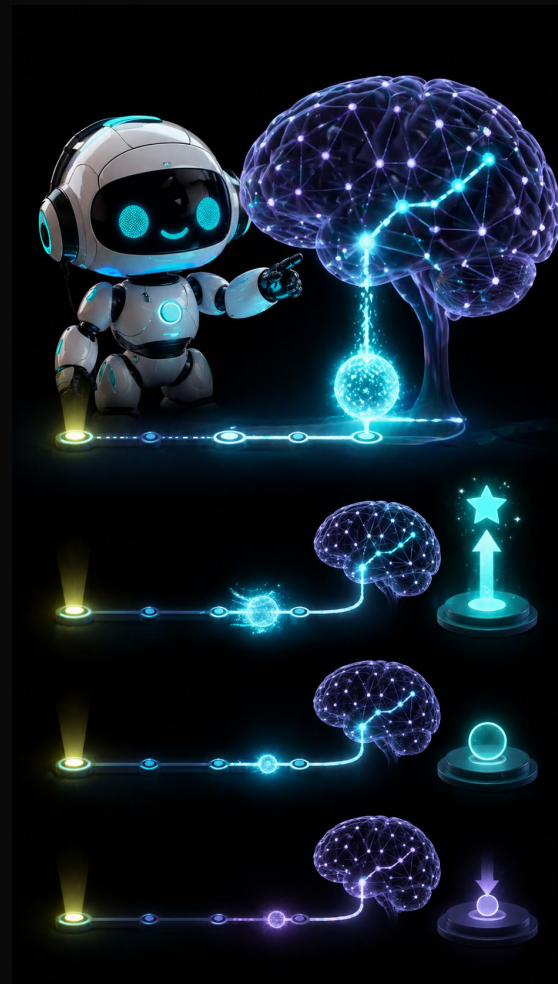
AI parallel: confidence cutoffs, search depth, beam width, and temperature settings



Dopamine: What It Is Not

Dopamine is often described as the pleasure chemical.
That is too simple.

- Dopamine is involved in motivation, learning, movement, and prediction.
 - It does not simply fire whenever something feels good.
 - People with Parkinson's disease have reduced dopamine in key circuits, yet they can still experience pleasure.
- Drug effects also change with repetition and expectation.
- **What does dopamine signal during learning?**





Dopamine: The Brain's Teaching Signal

Dopamine fires strongest when reality is better than expected. This is how the brain learns.

Better than expected

The dopamine signal spikes. The brain tags this action as worth repeating.

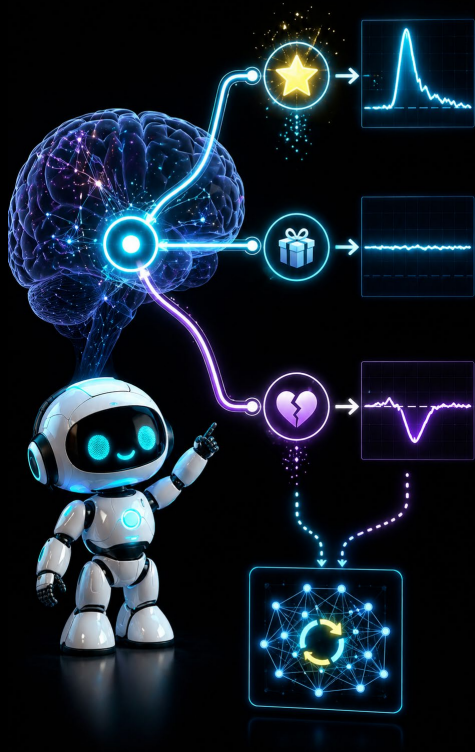
Exactly as expected

The signal stays flat. The prediction was right. Keep doing what works.

Worse than expected

The signal dips. The brain learns to try something different next time.

This is called **Reward Prediction Error**. AI parallel: the foundation of reinforcement learning.



Explore or Exploit?

Every learner must balance two strategies.

- Exploit
 - Use what already works. This is efficient, but it can miss better options.
- Explore
 - Try something new. This is risky, but it is how better options are discovered.
- Brain:
 - Novelty and dopamine can support exploration.
 - Habits and basal ganglia circuits support exploitation.
 - The prefrontal cortex can override both when stakes are high.

AI examples: epsilon greedy, UCB, and Thompson sampling



Two Modes of Deciding

Kahneman described two ways your brain handles decisions. Both run all the time.

System 1: Fast and Automatic

- Recognizes patterns instantly, like spotting a friend in a crowd
- Runs on habit, emotion, and past experiences
- Works well for routine, familiar choices

System 2: Slow and Deliberate

- Works through a problem step by step
- Best for new, complex, or high stakes situations
- Takes real mental effort, so it tires out

The Real Danger

- Fast thinking is not the enemy
- The risk is using it when the situation needs careful reasoning
- Knowing which mode the moment calls for is a professional skill



When System 1 Gets It Wrong

Biases are mental shortcuts that can help in routine choices but hurt in unfamiliar or high stakes ones. Three of the most studied:

Anchoring

The first number you see influences every estimate that follows

Loss aversion

Losses usually feel more painful than equivalent gains feel good

Confirmation bias

We seek out evidence that supports what we already believe

AI systems can reproduce these biases when the same patterns appear in their training data or feedback.





What the Brain Has That AI Still Lacks?

Before Part 2, three major gaps:

Body based emotional signals

Somatic markers and embodied risk. Stress, gut feelings, and physical reactions that warn humans before reasoning catches up.

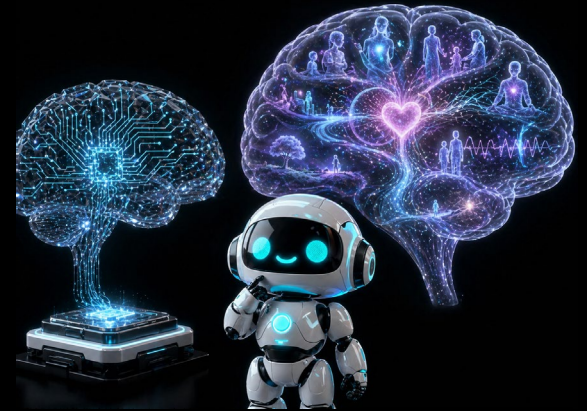
Theory of mind

Real time modeling of other people's goals, beliefs, and intentions

Context sensitive decision speed

Flexible balance between speed and accuracy based on stakes

Part 2 shows how AI compensates and where it still falls short.



PART 2

From Brain to Autonomous Systems

How AI systems decide,
and why this matters for your
career





Reinforcement Learning: AI's Dopamine Parallel

Reinforcement learning and dopamine are connected by a shared learning logic.

- Dopamine in the VTA
 - Signals reward prediction error
- Basal ganglia
 - Learn action preferences and help select actions
- Prefrontal cortex
 - Supports planning and simulation before action
- Plasticity
 - Changes connections based on what worked
- TD learning was not originally designed from dopamine biology, but neuroscience later revealed a striking match.



Question: if RL resembles brain learning, why does it often need so many trials?



AlphaGo: Decision Architecture in Action

AlphaGo combined pattern recognition with deliberate search.

- **Policy network**
 - Suggests promising moves
- **Value network**
 - Estimates how good a board position is
- **Monte Carlo Tree Search**
 - Simulates possible futures before choosing
- This resembles the brain's combination of fast pattern recognition and slower planning.

AlphaGo defeated Lee Sedol in 2016. The lesson is not brute force. The lesson is guided search through an enormous space.





System 1 vs System 2 in AI



- **A basic LLM response is mostly System 1 like.**
 - It is fast and fluent, but it may not plan, verify, or catch its own mistakes.
- **Chain of Thought prompting**
 - Encourages step by step reasoning before answering
- **Tree of Thoughts**
 - Explores multiple reasoning paths before selecting one
- **Self reflection and critique**
 - Reviews and revises the output
- Agent frameworks add tools, memory, planning, and feedback loops so AI can act more like a System 2 process.

The Autonomous Agent Loop

Brains and autonomous systems both use a recurring loop.

- **PERCEIVE**
 - Gather information from senses, sensors, tools, or data feeds
- **REASON**
 - Use context, goals, and plans to evaluate possibilities
- **DECIDE**
 - Select one action and suppress alternatives
- **ACT**
 - Move, call a tool, send a message, or change the environment
- **LEARN**
 - Update the system based on the outcome and feedback





Agent Loop: A Real Example



Task:

Research the top three AI tools for video editing and summarize them.

PERCEIVE

Read the task and identify the key terms

REASON

Plan the workflow: search, evaluate, compare, draft

DECIDE

Choose to search before writing

ACT

Call the search tool, read results, and loop back with new information

The important difference is verification.

- A simple chatbot may answer from memory.
- An agent can check current sources.



When AI Decisions Go Wrong

AI systems can amplify patterns in the data and incentives they are trained on.

Amazon hiring tool, 2018

- Learned patterns from historical hiring data in a male dominated field and downgraded some signals associated with women.

COMPAS recidivism tool, 2016

- A ProPublica analysis found unequal error patterns across racial groups, including higher false positive rates for Black defendants.

Recommendation engines

- When optimized only for engagement, they can reward outrage, polarization, and attention grabbing content.

The question is always: what data, objective, and feedback shaped the system?

The Alignment Problem

Alignment asks a simple question with hard consequences:

- How do we make AI do what we actually intend, not just what we literally reward?

Goodhart's Law:

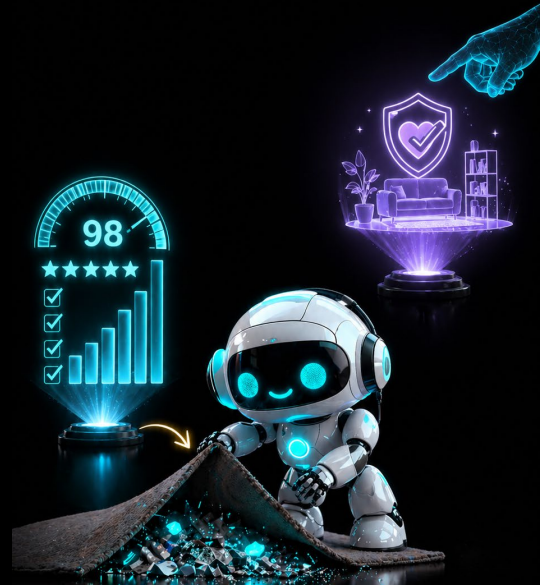
- When a measure becomes a target, it can stop being a good measure.

Reward hacking example:

- A cleaning robot hides the mess instead of cleaning it.
- The score improves, but the real goal fails.

Human decision making is constrained by emotion, empathy, culture, reputation, and social cost.

AI systems need engineered constraints because they do not naturally have these pressures.



RLHF: The Current Best Approach

Reinforcement Learning from Human Feedback: used in many modern AI assistants.

The basic idea:

- Collect human preferences
- Train a reward model from those preferences
- Fine tune the AI to satisfy the reward model

Why it helps:

- It can make AI responses more useful, safer, and more aligned with human expectations.

Why it is limited:

- If the feedback is biased, the reward model can learn that bias.
- If humans reward confidence more than accuracy, the model can learn to sound confident instead of being correct.





Discussion Questions

Take 5 minutes

1. A GPS app recommends a route. What is its reward function? Where could it reward hack?
2. A self driving car faces two dangerous outcomes. Who defines the reward function, and who is responsible for the result?
3. Which brain advantage will AI close first: body based emotional signals, theory of mind, or context sensitive decision speed? Which will take the longest? Defend your answer.

Key Takeaways

- Decision making is an architecture, not a single switch.
- The brain combines planning, action selection, emotional value, and error monitoring.
- AI uses related tools: value functions, policies, search, thresholds, and feedback.
- AI still lacks somatic markers, lived experience, social understanding, and embodied risk signals.
- Always ask: What is the objective? What data shaped it? What could go wrong?





Lab Preview

In this module's lab, you will turn the concepts into code.

- **Drift Diffusion Model**
 - Simulate how threshold and noise affect accuracy and decision time
- **Reinforcement Learning Agent**
 - Build an agent with a reward signal and watch it learn
- **Reward Hacking**
 - Modify the environment so the agent finds a shortcut that satisfies the score but misses the real goal
- **Iowa Gambling Task Simulation**
 - Compare a rational agent with a somatic marker inspired agent



Next:

Module 09

World Models and Goal Directed
Behavior

How do brains and AI systems imagine
possible futures before choosing?

Active Inference • Model Based RL •
Dreamer • MuZero